2. (a) Suppose two names from the same industry k

$$X_i^k = \sqrt{\rho_1 - \rho_0}\, X_k + \sqrt{\rho_0}\, X_M + \sqrt{1-\rho_1}\, \varepsilon_i$$

$$X_j^k = \sqrt{\rho_1 - \rho_0}\, X_k + \sqrt{\rho_0}\, X_M + \sqrt{1-\rho_1}\, \varepsilon_j$$

$$\mathrm{Cov}(X_i^k, X_j^k) = \mathrm{Corr}(X_i^k, X_j^k)$$

$$= (\rho_1 - \rho_0) + \rho_0 = \rho_1$$

If they are from different industries k_1 and k_2

$$X_i^{k_1} = \sqrt{\rho_1 - \rho_0}\, X_{k_1} + \sqrt{\rho_0}\, X_M + \sqrt{1-\rho_1}\, \varepsilon_i$$

$$X_j^{k_2} = \sqrt{\rho_1 - \rho_0}\, X_{k_2} + \sqrt{\rho_0}\, X_M + \sqrt{1-\rho_1}\, \varepsilon_j$$

$$\mathrm{Corr}(X_i^{k_1}, X_j^{k_2})$$

$= \rho_0$ (as all cross terms are 0)